

Q: /

$$(a) \quad S = \frac{10 \mu m}{5 \mu m} = 2$$

(2M)

$$I_{PML} \propto \frac{C_L}{S} \frac{V_{DD}}{S E_{ox}} = \frac{I_{PML}}{S^2}$$

$$f' = S^2 f = 2^2 \times 100 = 400 \text{ MHz}$$

(2M)

$$P' = S \times P = 2 \times 10 = 20 \text{ W}$$

(2M)

$$(b) \quad V = \frac{2V}{1V} = 2 \quad - (2M)$$

$$I_{PML}' \propto V I_{PML}$$

$$f' = \frac{S^2}{V} f = 2 \times 100 \text{ MHz} = 200 \text{ MHz} \quad - (3M)$$

$$P' = \frac{S}{V^3} P = \frac{10 \text{ W}}{2^2} = 2.5 \text{ W} \quad - (3M)$$

$$(c) \quad P' = \frac{S}{V^3} P$$

$$P' = 5 \text{ W}, \quad P = 10 \text{ W}$$

$$\therefore P'/P = 0.5$$

$$S/V^3 = 0.5$$

$$V = 1.587 \quad -$$

(2M)

$$\therefore V = 1.26 \text{ V} \quad -$$

(2M)

$$\begin{aligned} f' &= \frac{S^2}{V} f = 2.52 f \\ &= 252 \text{ MHz} \end{aligned}$$

(2M)

Q2 $\Rightarrow$ $V_{OH} = V_{DD}$ For $V_{OL} \Rightarrow I_H(\text{linear}) = I_P(\text{sat})$

$$\frac{\mu_n C_{ox}}{2} \left(\frac{W}{L} \right)_n \left[2(V_{in} - V_{th_n})V_{OL} - V_{OL}^2 \right] = \frac{\mu_p C_{ox}}{2} \left(\frac{W}{L} \right)_p (V_{in} - V_{DD} - V_{th_p})^2$$

solving

$$V_{OL} \approx 5 \text{ mV}$$

$\rightarrow$ 4H

b) $C_L = 0.95 \text{ pF} = 0.95 \text{ pF}$

avg current method $\tau = \frac{C \Delta V}{I_{avg}}$

$$\Delta V = 2.5 = V_{DD}/2$$

i) $I_{avg} = \frac{1.6 \text{ nA} + 1.375 \text{ nA}}{2}$ 7 2

ii) $I_{avg} = 14.875 \text{ nA}$

$$\tau_{PLH} = \frac{0.95 \text{ pF} \times 2.5 \text{ V}}{2(1.6 \text{ nA} + 1.375 \text{ nA})}$$

$$\tau_{PLH} = 2.017 \text{ ns}$$

$$1.596 \text{ ns}$$

$\rightarrow$ 3H

$$\tau_{PHL} = \frac{0.95 \times 2.5}{14.875}$$

$$\tau_{PHL} = 0.159$$

(Alternative)

b) $R_{eq} = \left(\frac{5}{1.6 \text{ nA}} + \frac{2.5}{1.375 \text{ nA}} \right) \div 2 = 2.467 \text{ k}$

$R_{eq} = \left(\frac{5}{16 \text{ nA}} + \frac{2.5}{13.75 \text{ nA}} \right) \div 2 = 0.246$

$$\tau = 0.69 R_{eq} C_L = 1.617 \text{ ns}$$

$$\tau = 0.161 \text{ ns}$$

c) $P = C_{tot} \times V_{DD}^2 \times f = 0.95 \text{ pF} \times 5^2 \times 100 \text{ MHz}$

$$P \approx 2.3 \text{ mW}$$

$\rightarrow$ 3H

$V_{in} < V_{Th}$, Load $\Rightarrow$ Saturation
Driver $\Rightarrow$ Cut off.

$$\text{So, } \frac{K_{Load}}{2} \left[(V_{DD} - V_{out} - V_{Th}) \right]^2 = 0$$

$$V_{out} = V_{OH}, \quad V_{out} = V_{OH} = V_{DD} - V_{Th} = 2.5 - 0.4 = \boxed{2.1 \text{ V}}$$

$V_{IL} \geq V_{Th}$ Load: Saturation
Driver: Saturation.

$$\frac{K_{Load}}{2} (V_{DD} - V_{out} - V_{Th})^2 = \frac{K_{Driver}}{2} (V_{in} - V_{Th})^2$$

$$\Rightarrow (2.1 - V_{out})^2 = 8 (V_{in} - V_{Th})^2$$

$$\Rightarrow 2(2.1 - V_{out}) \left[-\frac{dV_{out}}{dV_{in}} \right] = 8 \times 2 (V_{in} - V_{Th})$$

$$\Rightarrow V_{out} = 5.3 - 8V_{in}$$

$$\text{Now, } \left[2.1 - (5.3 - 8V_{in}) \right]^2 = 8 (V_{in} - V_{Th})^2$$

$$V_{in} = V_{IL}, \quad 56V_{IL}^2 - 44.8V_{IL} + 8.96 = 0$$

$$\boxed{V_{IL} = 0.4 \text{ V}}$$

$V_{in} = V_{OH}$, Load: Saturation.
Driver: Linear.

$$\frac{K_{Load}}{2} \left(\overset{V_{DD} - V_{Th}}{\uparrow} (2.1 - V_{OL}) \right)^2 = \frac{K_{Driver}}{2} \left[2(V_{OH} - V_{Th})V_{OL} - V_{OL}^2 \right]$$

$$\Rightarrow (2.1 - V_{OL})^2 = 8 \left[2(2.1 - 0.4)V_{OL} - V_{OL}^2 \right]$$

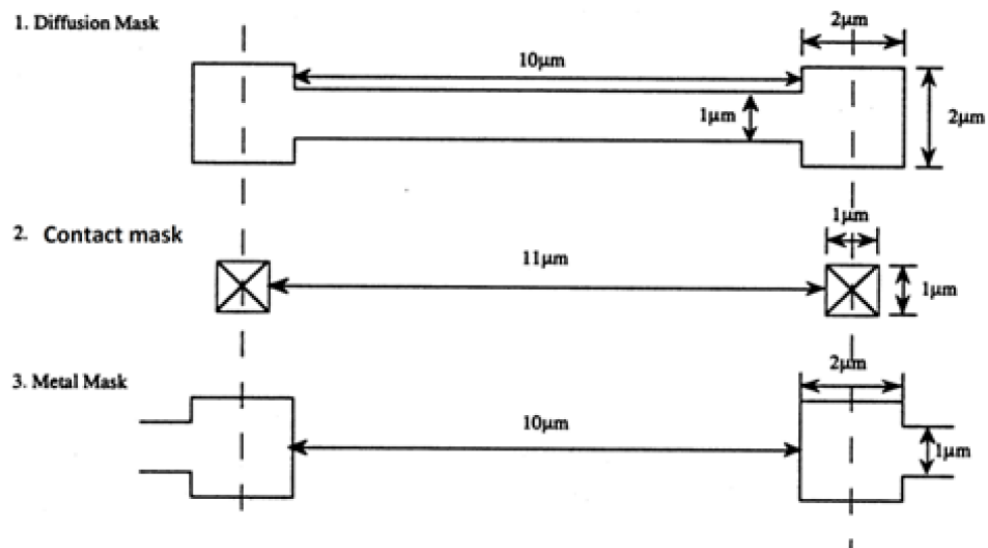
$$\Rightarrow 9V_{OL}^2 - 31.4V_{OL} + 4.41 = 0$$

$$V_{OL} = \boxed{0.144 \text{ V}}$$

$$NML = V_{IL} - V_{OL} = 0.4 - 0.144 = \boxed{0.256 \text{ V}}$$

Q4

Ans. $R = \rho L/A = \rho L/t \cdot w = R_s L/W$ and thus, $L = R \cdot W / R_s = 10 \mu\text{m}$.



a-----5

b-----3+2

c-----3+2

d-----3+2